

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Mathematics
2. **Subject Name:** Computer organization and architecture **L-T-P:** 3-0-0 **Credits:** 3
3. **Pre-requisites:** None
4. **Syllabus and reference books:**

Computer arithmetic, point representations, introduction to CISC processor architecture, instruction set and addressing modes. Hardware design principles, polling of processors [6 Lectures], Memory types and interfacing and timing I/O handling, interrupts and DMA and device interfaces-CRT, floppy disk, HDD, optical disk, serial interfaces and data acquisition [7 Lectures], Operating system concepts and architectural support- privileged mode, software interrupts, memory hierarchy and virtual memory, multi-processors concept [6 Lectures], Cache memory, pipe lining and introduction, RISC processors, super scalar processors. Resource scheduling problems-their solutions and analysis [7 Lectures], Storage management, secondary storage, virtual memory, segmentation and paging. Concurrent processes and their synchronization abstractions. Producer & consumer problems [7 Lectures], Mutual exclusion and dead-lock, properties of concurrent programs, critical section problem, semaphores, test and set, atomic operations, security and recovery [7 Lectures].

Reference Books:

1. Computer Organization and Design, D. A. Patterson and J. L. Hennessy.
2. Computer Architecture and Organization, J. P. Hayes.
3. Computer Architecture, B Parhami.

1. Lecture-wise break-up:

Sl. No.	Topic	No. of Lectures
1.	Computer arithmetic, point representations, introduction to CISC processor architecture, instruction set and addressing modes. Hardware design principles, polling of processors	6
2.	Memory types and interfacing and timing I/O handling, interrupts and DMA and device interfaces-CRT,	7

	floppy disk, HDD, optical disk, serial interfaces and data acquisition	
3.	Operating system concepts and architectural support- privileged mode, software interrupts, memory hierarchy and virtual memory, multi-processors concept	6
4.	Cache memory, pipe lining and introduction, RISC processors, super scalar processors. Resource scheduling problems-their solutions and analysis	7
5.	Storage management, secondary storage, virtual memory, segmentation and paging. Concurrent processes and their synchronization abstractions. Producer & consumer problems, Mutual exclusion and dead-lock, properties of concurrent programs, critical section problem, semaphores, test and set, atomic operations, security and recovery	14
Total number of hours		40